UCLouvain

École polytechnique de Louvain

ÉCOLE POLYTECHNIQUE DE LOUVAIN

LEPL2022 - Health Innovation Classes

Problem definition

March 5th, 2026

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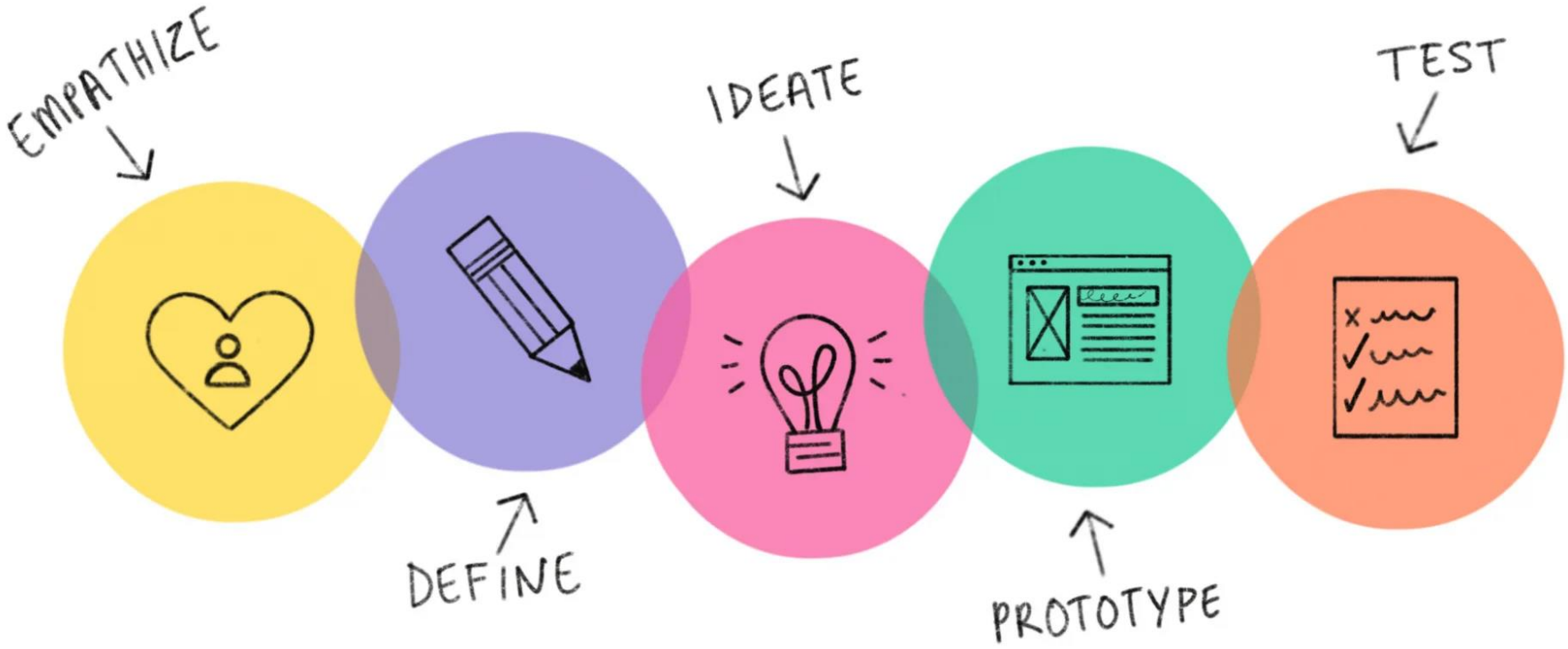
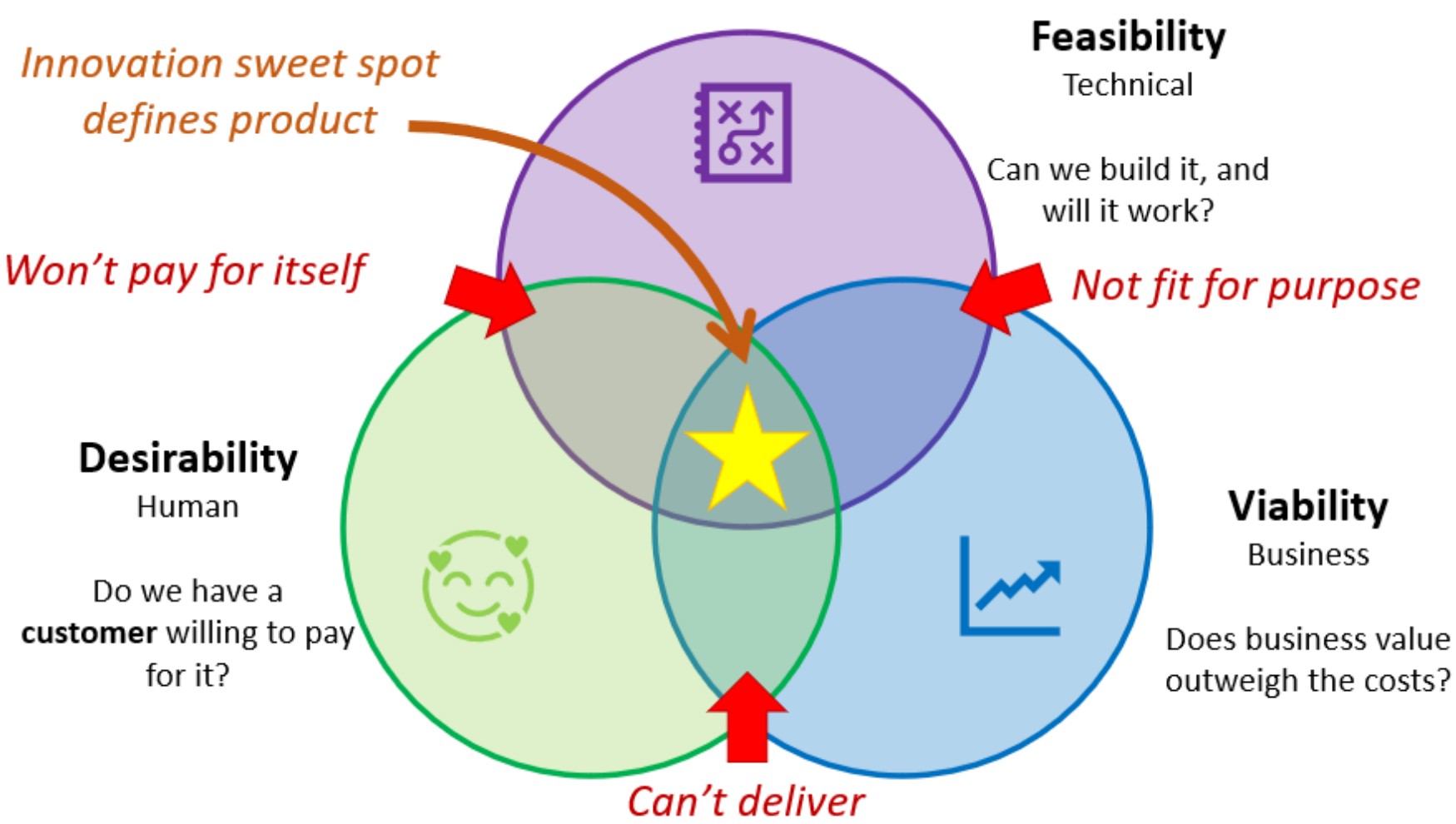
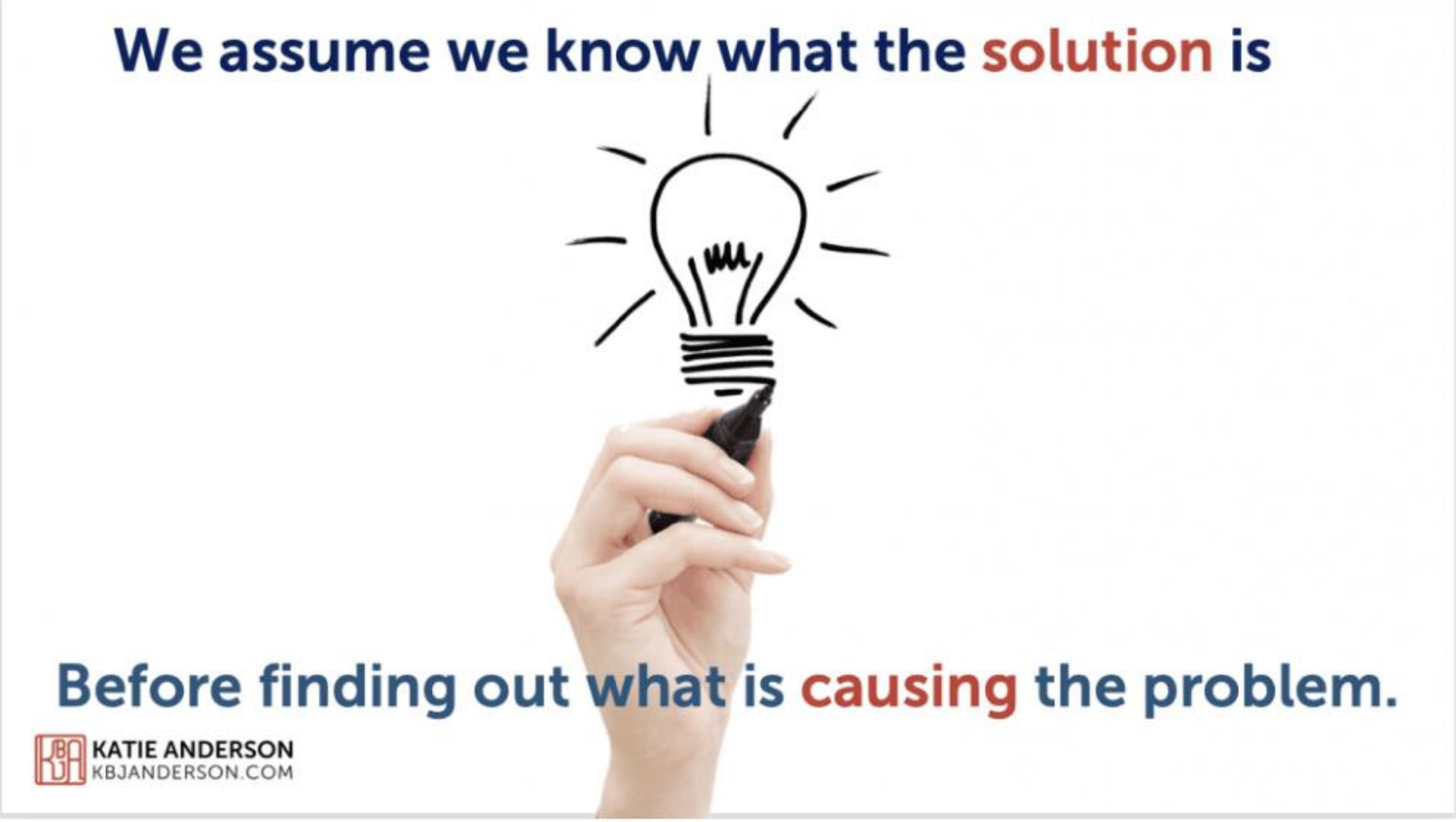
Reminder

DESIGN THINKING PROCESS

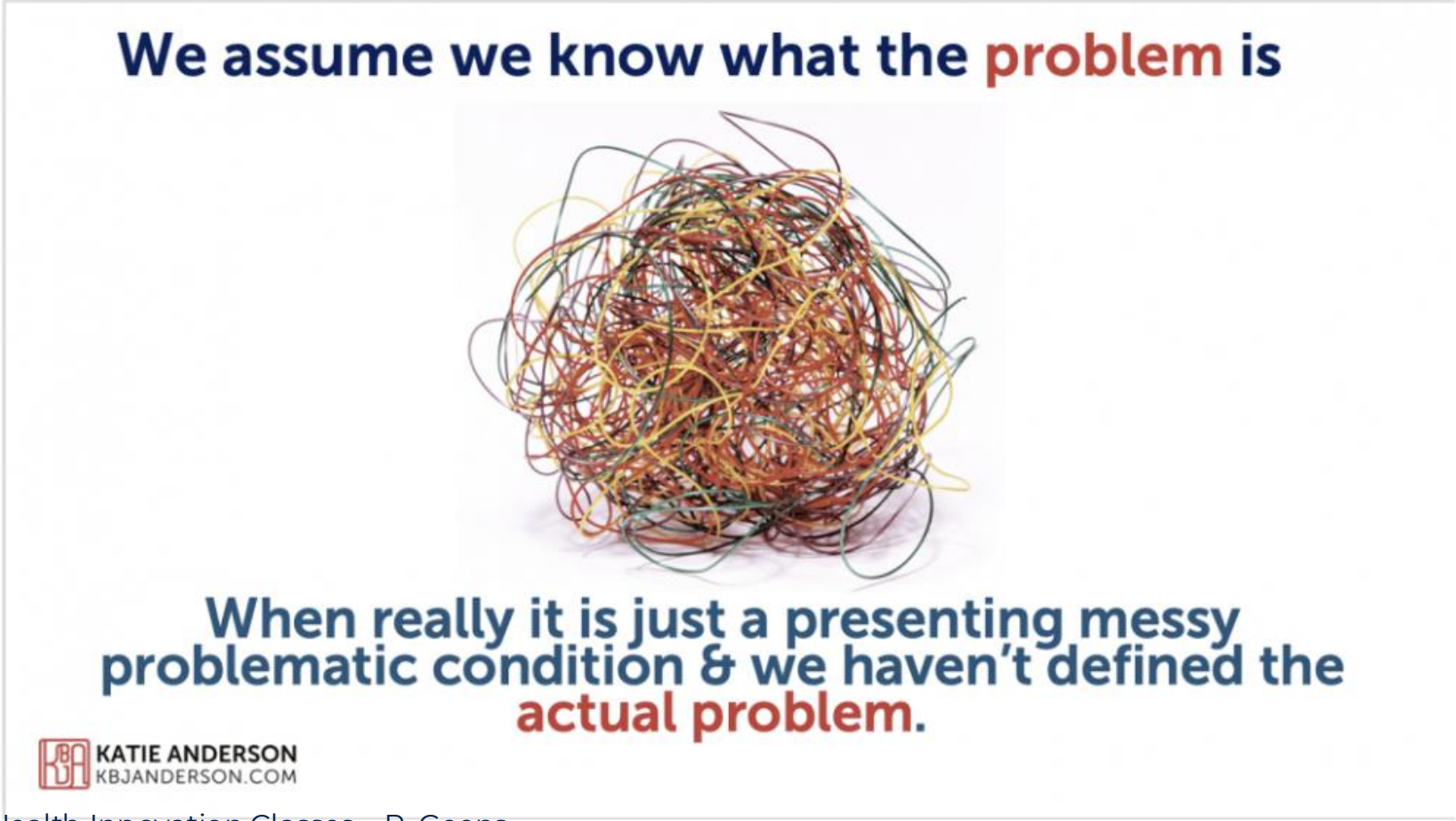



Innovation sweet spot defines product
Feasibility
Technical
Can we build it, and will it work?
Won't pay for itself
Not fit for purpose
Desirability
Human
Do we have a **customer** willing to pay for it?
Viability
Business
Does business value outweigh the costs?
Can't deliver

DEFINING THE PROBLEM BEFORE JUMPING ON SOLUTIONS



DEFINING THE **REAL PROBLEM** BEFORE JUMPING ON SOLUTIONS



From Field Insights to a Powerful Problem Statement

PROBLEM DEFINITION PRINCIPLES

- **Human-centered,** focused on users' needs rather than solutions
- **Grounded in evidence** from research, observation, and interviews
- **Systemic**, accounting for interrelations and interdependencies between observations (actors, processes, constraints, context)
- Targets **root causes** and insights, not just visible symptoms
- **Clearly scoped**: neither too broad nor too narrow
- Framed as **a need**, not as a user failure or lack
- **Open-ended**, allowing multiple possible solutions
- **Shared and aligned** across the team and stakeholders

Today’s Objective

BY THE END OF THE SESSION TODAY YOU WILL BE ABLE TO:

- **Choose the right problem-framing tool** depending on your specific project and progress within that project
- Use 6 **complementary** methodologies
- **Transform** your observations into a strong problem statement

6 METHODS WE WILL USE (amongst many others)

- Care pathway mapping
- 5 Why's
- Ishikawa
- Network of problems
- Casual loops
- How Might We

EACH METHOD AIMS AT ANSWERING A DIFFERENT QUESTION

Care pathway	• Where does the problem occur?
5 Why's	• What is the root cause?
Ishikawa	• What factors contribute to the root cause?
Network of problems	• How are problems interconnected?
Casual loop	• Why does the problem persist over time?
How Might We?	• How do we open the solution space?

Care Pathway Mapping

CARE PATHWAY MAPPING

Visualizing the real patient journey:

- Steps
- Actors
- Tools
- Emotions
- Pain points

Purpose:

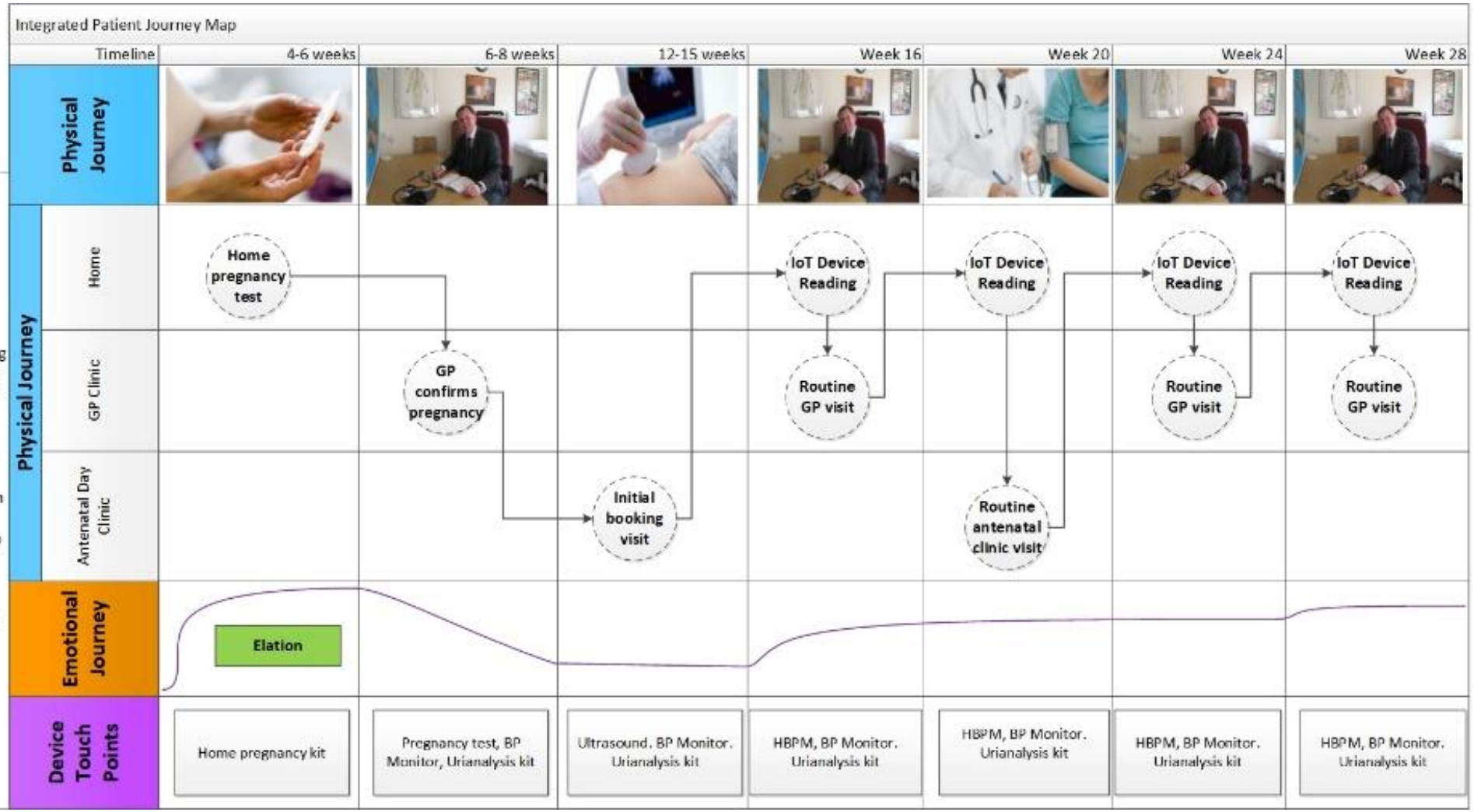
- Identify where innovation matters

Sheila

Age: 31
Occupation: Accountant

Risk profile: Sheila is at risk of hypertension during pregnancy. It is her first pregnancy and she has a family history of pre-eclampsia.

Summary: Based on her risk profile, Sheila is selected to partake in the clinical study. However, no complications arise during her pregnancy and therefore she follows the standard medical pathway.



EXERCISE – 5 MINS

Map the full journey from:

- leaving home → sitting in the classroom ready to learn

Include:

- steps
- actors
- tools
- emotions
- pain points

5 Why’s

5 Why's

Goal:

- Move from symptom → root cause

Rule:

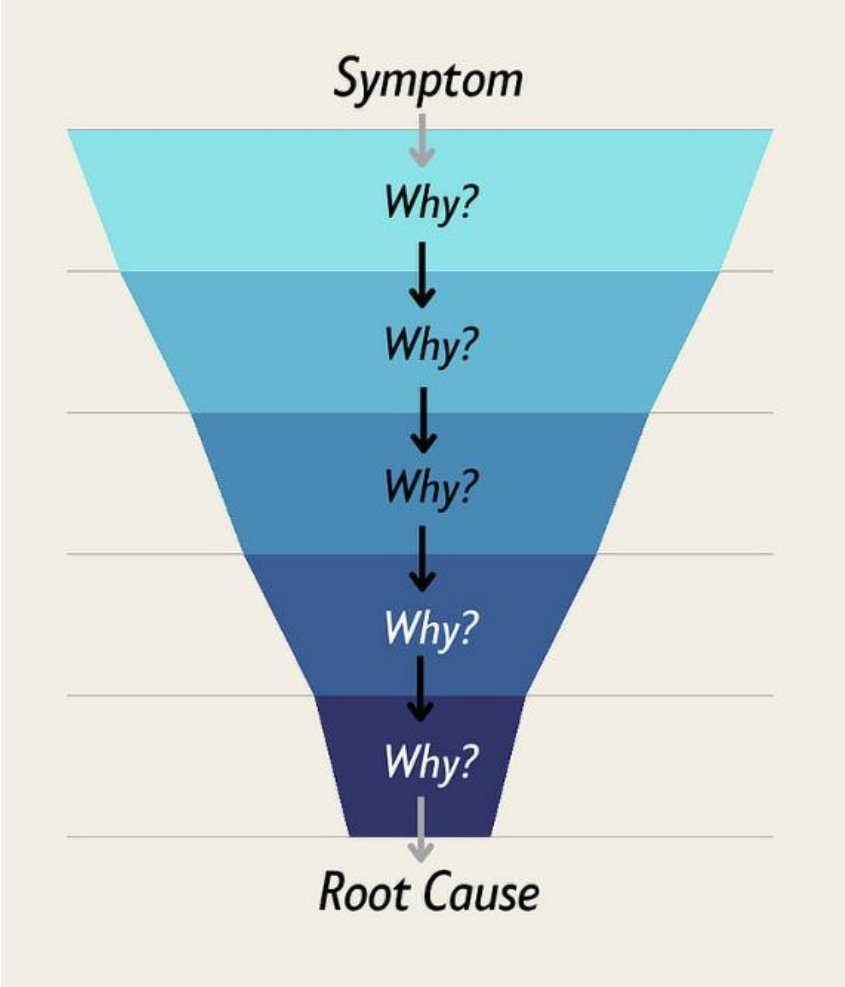
- Each answer must be factual and observable.

Method

- Start from one pain point in your pathway.
- Ask WHY five times.

Outcome:

- A root cause you can act on



EXERCISE – 5 MINS

From previous exercise, select one pain point

- Ask “Why is that?” 5 times
- See if you can identify a root cause you can act on

Ishikawa 石川
Fishbone diagram

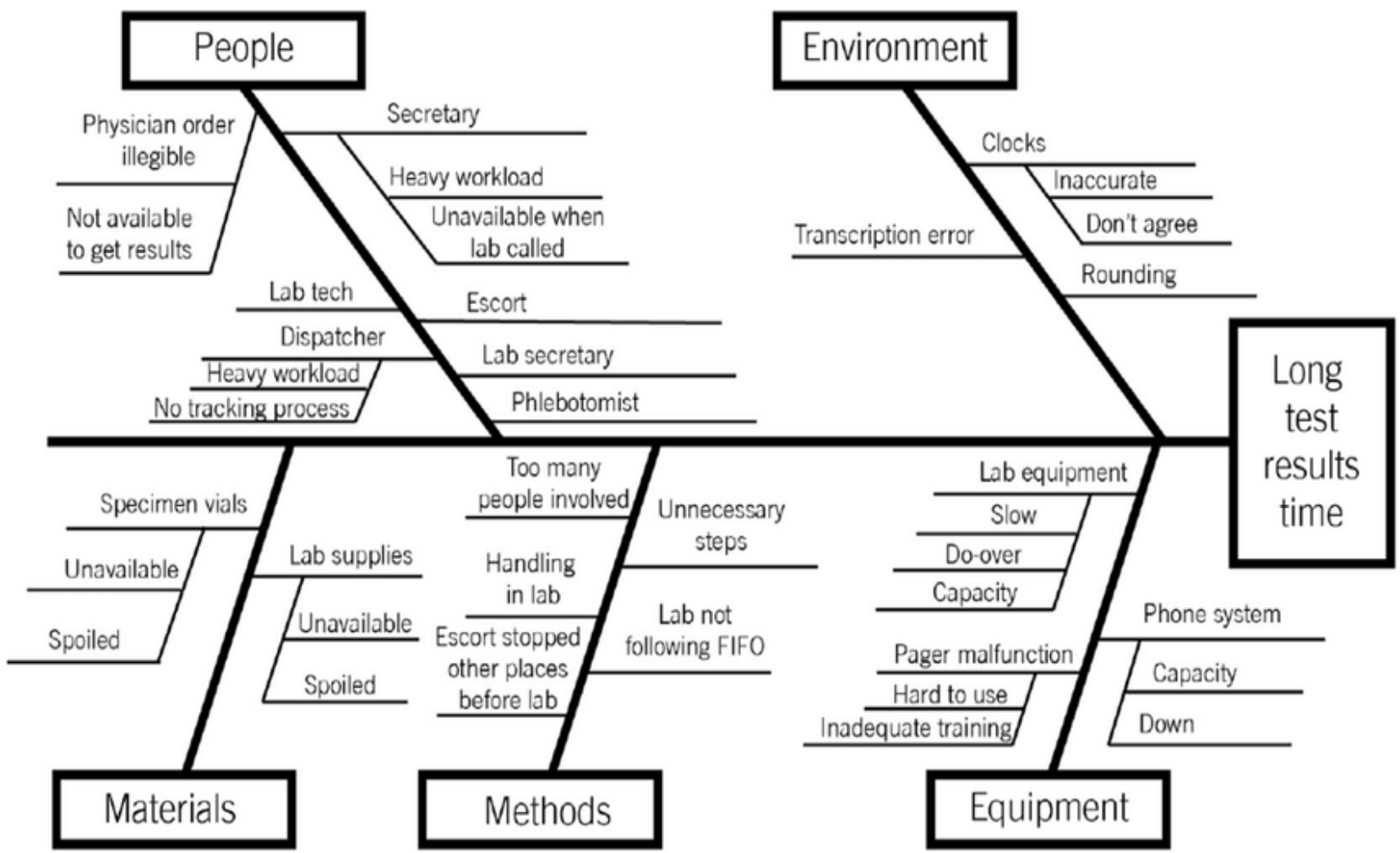
ISHIKAWA / FISHBONE DIAGRAM

A structured view of causes – usually 6 categories (the 6M) but can be simplified down to 5 (sometimes 4)

- Manpower (people)
- Methods (procedures, instructions, implicit ways of working)
- Materials (raw materials, disposables, tools)
- Machine (capital equipments, software)
- Mother nature (working environment, external conditions)
- Measure (measuring tools, reporting)

Can include sub-levels

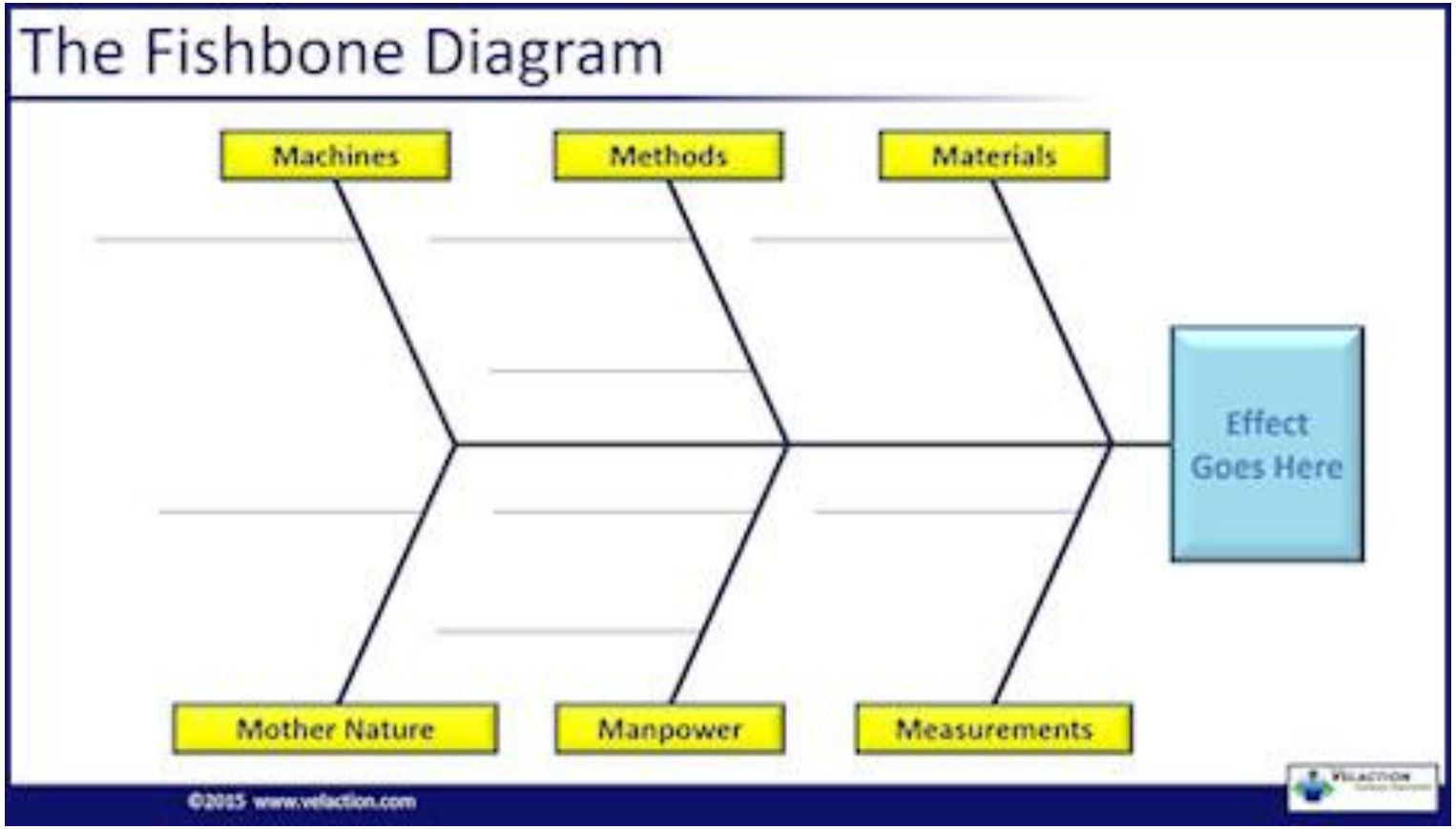
Shows the multifactorial nature of problems in a structured and visual manner.



Introduction to quality improvement tools for the clinician - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/An-Ishikawa-or-Fishbone-diagram-demonstrates-the-relationships-between-potental-causes-of_fig1_325345322 [accessed 2 Mar 2026]

EXERCISE – 5 MINS

Symptom: Dinner is always ready very late in our shared student flat



Network of problems & CLDs

In complex environments, problems are not isolated. Treating only one problem may have detrimental consequences on others (risk/benefit of the solution is not favourable) or even backfire (amplification mechanisms)

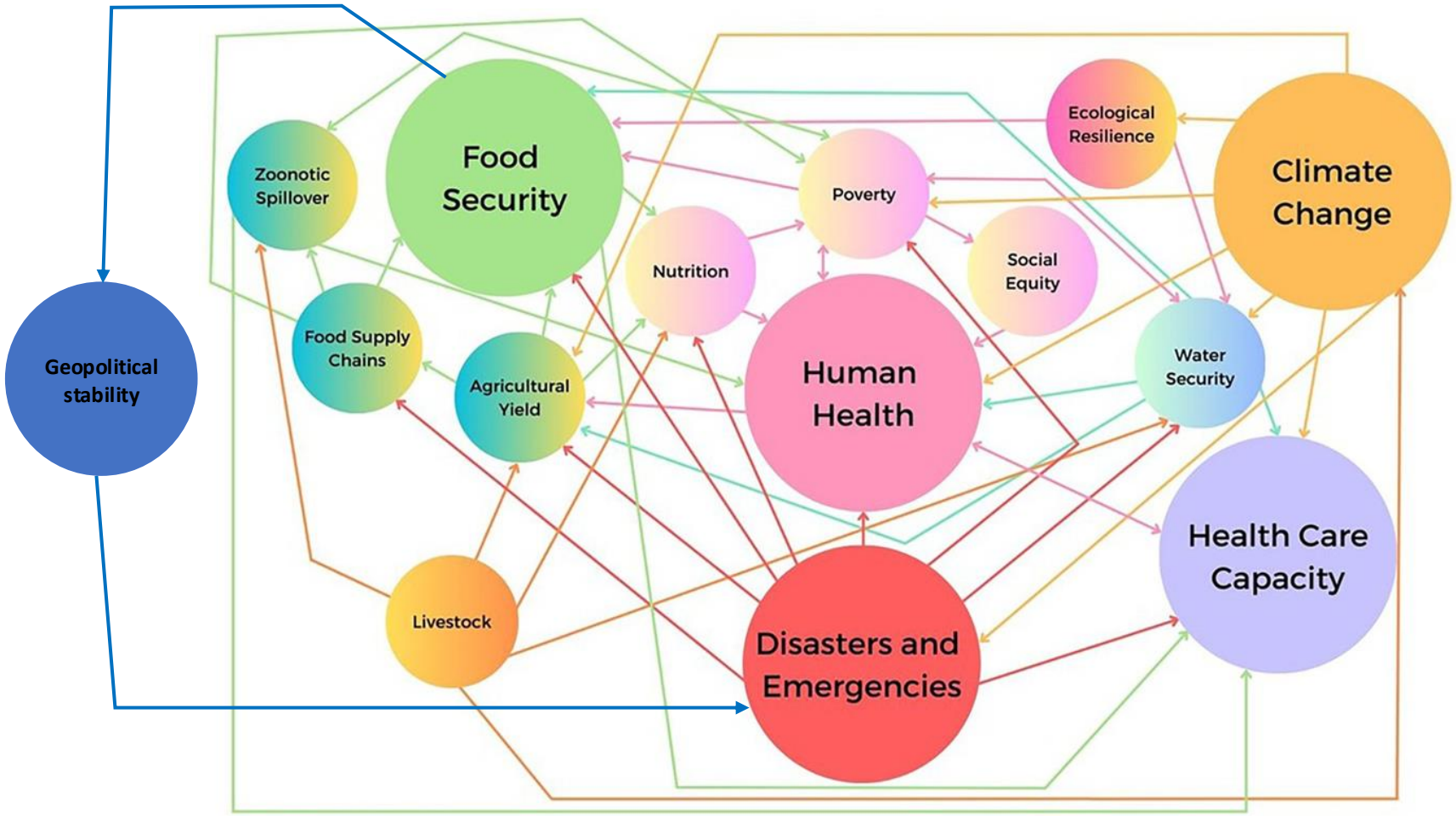
We need to map them

- Identify the root problems
- Adjacent, resulting problems and consequences
- Interdependencies

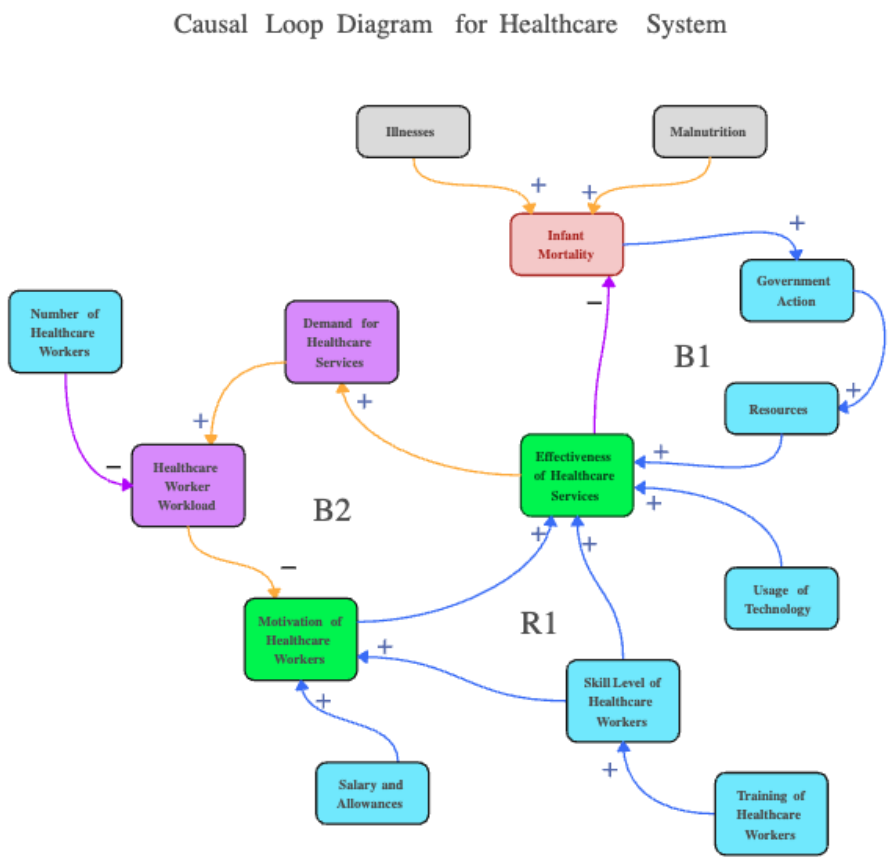
Purpose:

- Identify leverage points

NETWORK OF PROBLEMS



CASUAL LOOP DIAGRAMS



- Identify core parameters
- Identify other influencing parameters (Ishikawa may help). Keep it relatively simple
- Create relations
- A increases → B increases +
  A decreases → B decreases +
- A increases → B decreases -
  A decreases → B increases -
- Identifiy loops
  - + + + … → **R**einforcement = virtuous or vicious cycle
  - + - + … → **B**alancing
- Identify on which parameters and loops you can actually act on act to reduce problems/symptoms

How might we

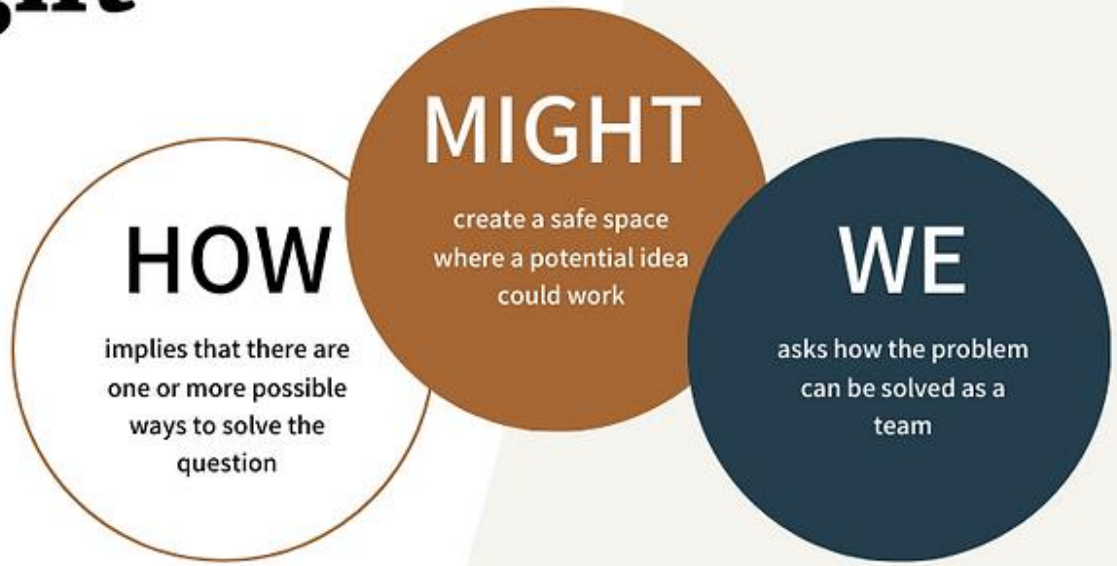
HOW MIGHT WE

At the end of problem analysis we usually have:

- root causes
- systemic understanding
- a clear problem statement

But a problem statement alone can constrain creativity.

How Might We



PROJECT TAILWIND

HOW MIGHT WE - EXERCISE

Observation in an Emergency Department

During several busy shifts, nurses frequently need to locate specific medical equipment (e.g., infusion pumps, monitoring devices) for urgent patient care.

However:

- Equipment is stored in multiple locations
- Some devices are already in use
- Others are misplaced or not ready (e.g., low battery)

As a result, nurses often spend **several minutes searching for the right equipment**, while:

- doctors are waiting
- patients require immediate attention
- stress in the team increases

Nurses waste time searching for equipment

A fancy tracking system!

Oh, an App!!

RFID tags!!!!

How might we…

help emergency nurses access critical equipment in less than 30 seconds?

Summary

Method	How it works	Best Used When…	Not Appropriate When…
Care Pathway Mapping	Visual representation of the steps a user goes through in a process (e.g., patient journey). Helps locate pain points, delays, and breakdowns in real contexts.	When you want to understand **where problems occur in real life** and how different actors interact across a process. Very useful for service design and healthcare workflows.	When the project does not involve a **clear process or journey**, or when you already know exactly where the problem occurs and need deeper causal analysis.
5 Whys	Iterative questioning technique to identify the **root cause** of a problem by repeatedly asking "Why?".	When investigating a **specific observable problem** and trying to move beyond symptoms to underlying causes.	When the problem is **highly systemic with multiple interacting causes**, which may oversimplify reality into a single chain.
Ishikawa (Fishbone Diagram)	Structured diagram organizing possible causes of a problem into categories (people, process, technology, environment, organization).	When analyzing **multifactorial problems** and needing a structured overview of contributing factors.	When trying to understand **dynamic interactions or feedback loops over time**, which the method does not capture.
Network of Problems	Mapping of interconnected problems showing relationships between root causes and consequences.	When problems involve **multiple stakeholders or cascading effects**, and you want to identify leverage points.	When you need **precise causal analysis of one problem**, or when the scope is too large and the map becomes unclear.
Causal Loop Diagram (CLD)	System dynamics tool showing feedback loops between variables. Reveals **reinforcing and balancing mechanisms** explaining why problems persist.	When analyzing **complex systems evolving over time**, such as organizational behavior, habits, resource dynamics, or healthcare systems.	When the problem is **very local or technical**, or when the team lacks enough understanding to identify meaningful variables.
How Might We (HMW)	Open-ended question that reframes a problem into an **innovation opportunity**, keeping the solution space open.	At the **end of problem definition**, when you want to transition from analysis to ideation.	When the problem is **not yet clearly understood**, which would lead to superficial or misdirected ideas.

Questions

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